

CS57300
PURDUE UNIVERSITY

DATA MINING

OVERVIEW

- ▶ Task specification
- ▶ Knowledge representation
- ▶ Learning technique
 - ▶ Search + Evaluation
- ▶ **Prediction and/or interpretation**

INFERENCE AND INTERPRETATION

- ▶ Prediction technique
 - ▶ Method to apply learned model to new data for prediction/analysis
 - ▶ Only applicable for predictive and some descriptive models
 - ▶ Prediction is often used during **learning** (i.e., search) to determine value of scoring function
- ▶ Interpretation of results
 - ▶ Objective: significance measures
 - ▶ Subjective: importance, interestingness, novelty

EXAMPLE: IDENTIFYING EMAIL SPAM

- ▶ Task
 - ▶ Design automatic spam detector that can differentiate between labeled emails
 - ▶ Data
 - ▶ Table of relative word/punctuation frequencies
 - ▶ Knowledge representation
 - ▶ If/then rules with conjunctions of features
 - ▶ Learning technique
 - ▶ **Search** over set of rules, **select** rule with maximum accuracy on training data
- TABLE 1.1.** Average percentage of words or characters in an email equal to the indicated word or character. We have chosen the words and characters showing the largest difference between **spam** and **email**.

	george	you	your	hp	free	hpl	!	our	re	ed
spam	0.00	2.26	1.38	0.02	0.52	0.01	0.51	0.51	0.13	0.0
email	1.27	1.27	0.44	0.90	0.07	0.43	0.11	0.18	0.42	0.2

if (%george < 0.6) & (%you > 1.5) then sp
else em

TABLE 1.1. Average percentage of words or characters in an email message equal to the indicated word or character. We have chosen the words and characters showing the largest difference between **spam** and **email**.

	george	you	your	hp	free	hpl	!	our	re	edu	remove
spam	0.00	2.26	1.38	0.02	0.52	0.01	0.51	0.51	0.13	0.01	0.28
email	1.27	1.27	0.44	0.90	0.07	0.43	0.11	0.18	0.42	0.29	0.01

```
if (%george < 0.6) & (%you > 1.5)    then spam
                                     else email.
```

EXPLORATORY DATA ANALYSIS

EXPLORATORY DATA ANALYSIS

- ▶ Data analysis approach that employs a number of (mostly graphical) techniques to:
 - ▶ Maximize insight into data
 - ▶ Uncover underlying structure
 - ▶ Identify important variables
 - ▶ Detect outliers and anomalies
 - ▶ Test underlying modeling assumptions
 - ▶ Develop parsimonious models
 - ▶ Generate hypotheses from data

EXPLORATORY DATA ANALYSIS

- ▶ Low-dimensional data
 - ▶ Summarizing data with simple statistics
 - ▶ Plotting raw data (1D, 2D, 3D)
- ▶ Higher-dimensional data
 - ▶ Dimensionality reduction: e.g., principal component analysis

DATA VISUALIZATION

VISUALIZATION

- ▶ Human eye/brain have evolved powerful methods to detect structure in nature
- ▶ Display data in ways that exploit human pattern recognition abilities
- ▶ Limitation: Can be difficult to apply if data size (number of dimensions or instances) is large

DATA SUMMARIZATION

- ▶ Measures of location

- ▶ Mean: $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x(i)$
- ▶ Median: value with 50% of points above and below
- ▶ Quartile: value with 25% (75%) points below
- ▶ Mode: most common value

DATA SUMMARIZATION

- ▶ Measures of dispersion or variability

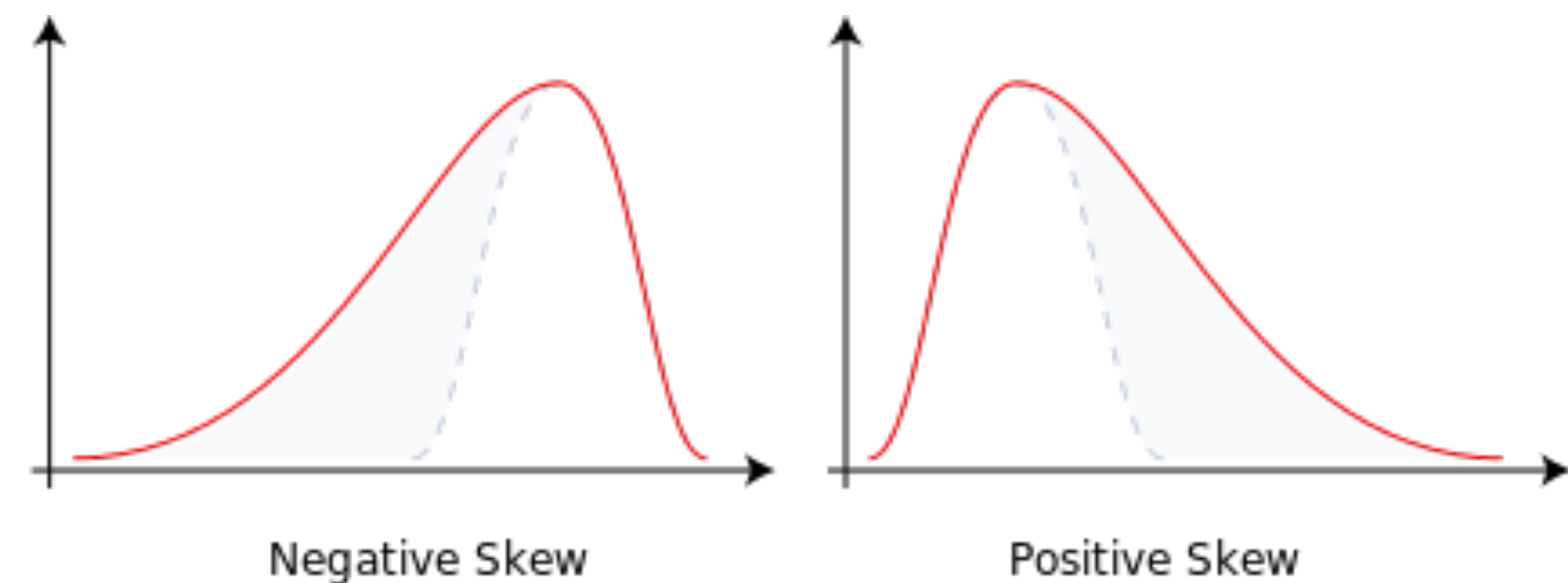
- ▶ Variance: $\hat{\sigma}_k^2 = \frac{1}{n} \sum_{i=1}^n (x(i) - \mu)^2$

- ▶ Standard deviation: $\hat{\sigma}_k = \sqrt{\frac{1}{n} \sum_{i=1}^n (x(i) - \mu)^2}$

- ▶ Range: difference between max and min point

- ▶ Interquartile range: difference between 1st and 3rd Q

- ▶ Skew: $\frac{\sum_{i=1}^n (x(i) - \hat{\mu})^3}{(\sum_{i=1}^n (x(i) - \hat{\mu})^2)^{\frac{3}{2}}}$

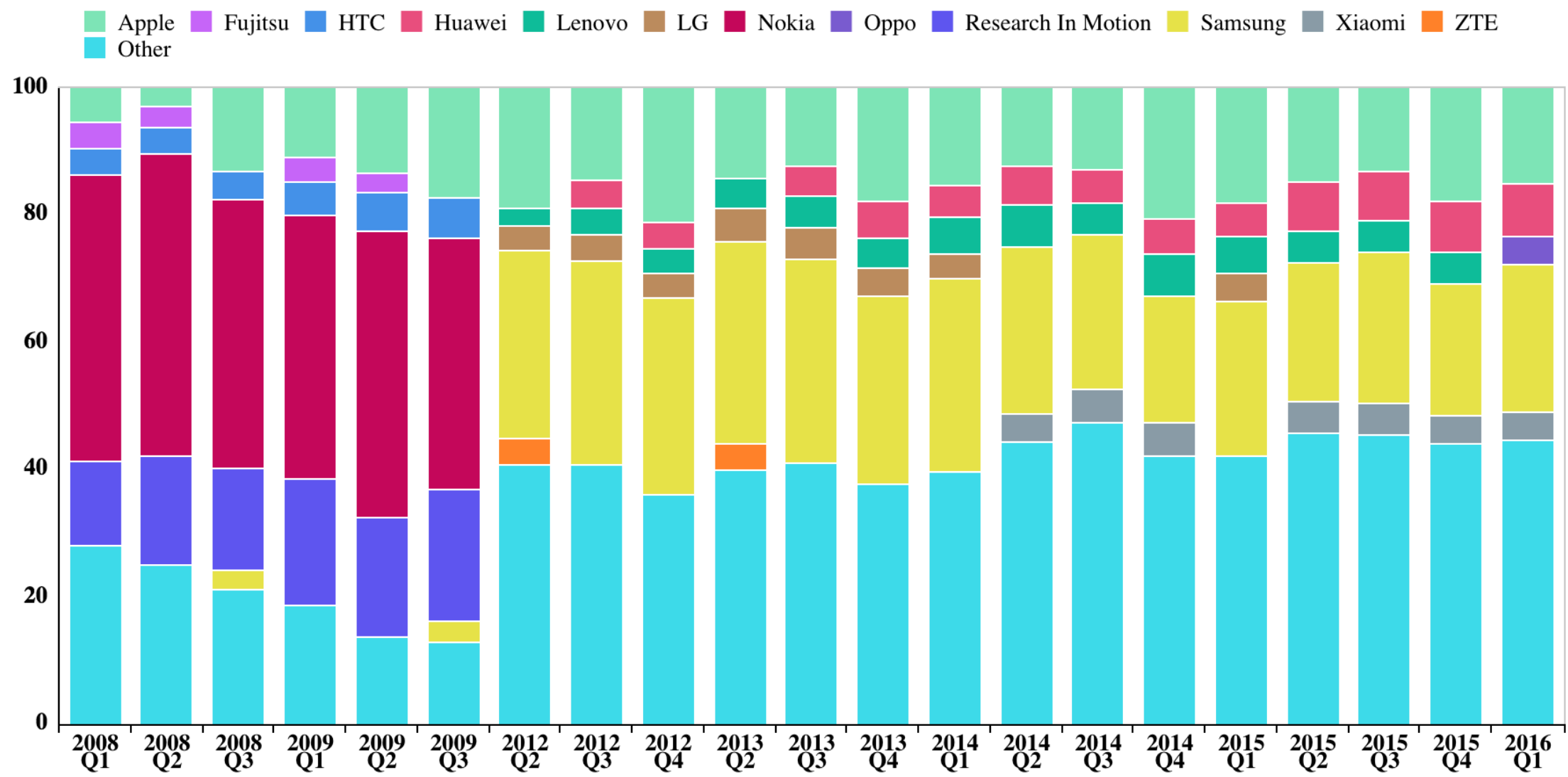
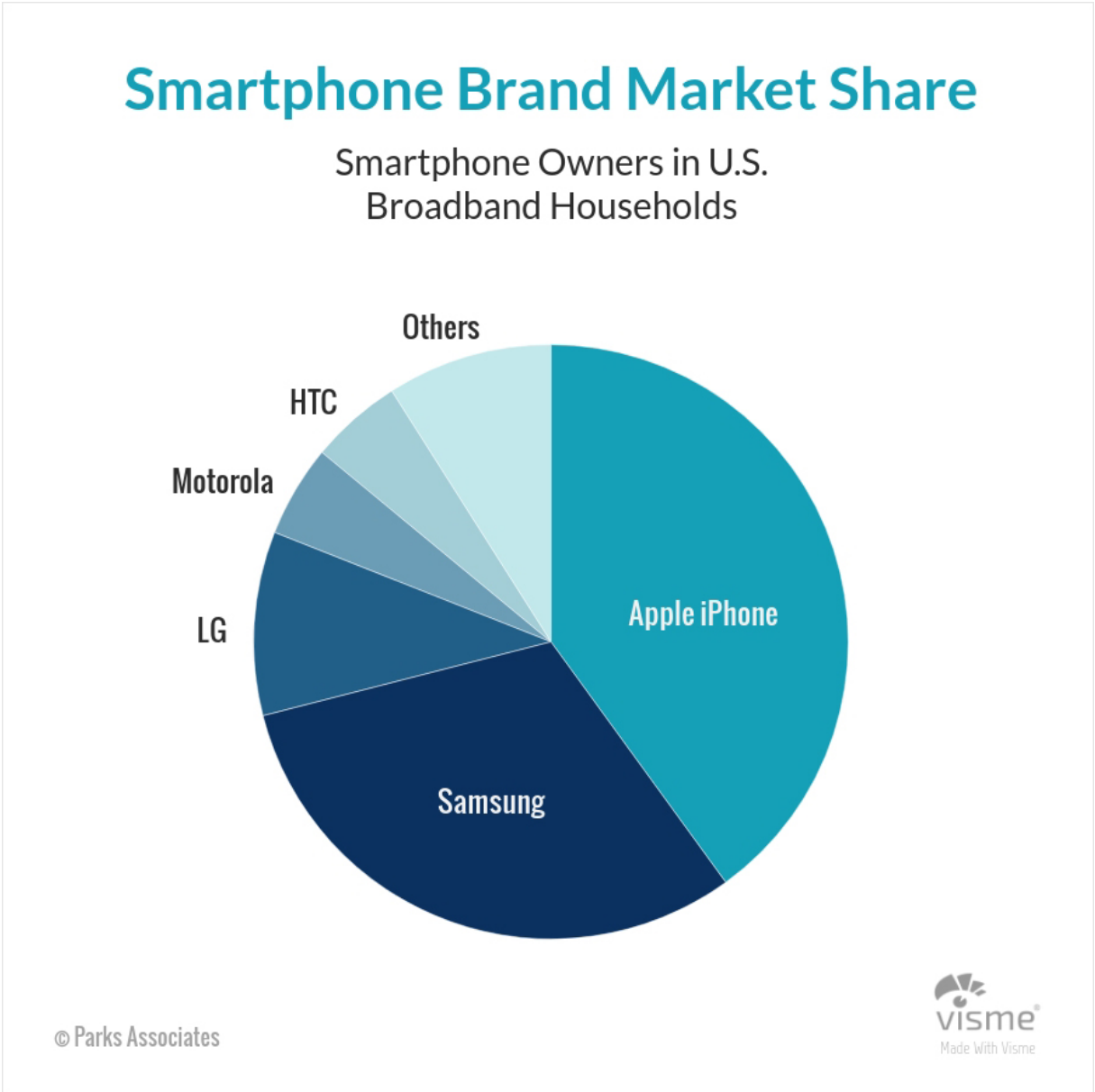


DATA VISUALIZATION

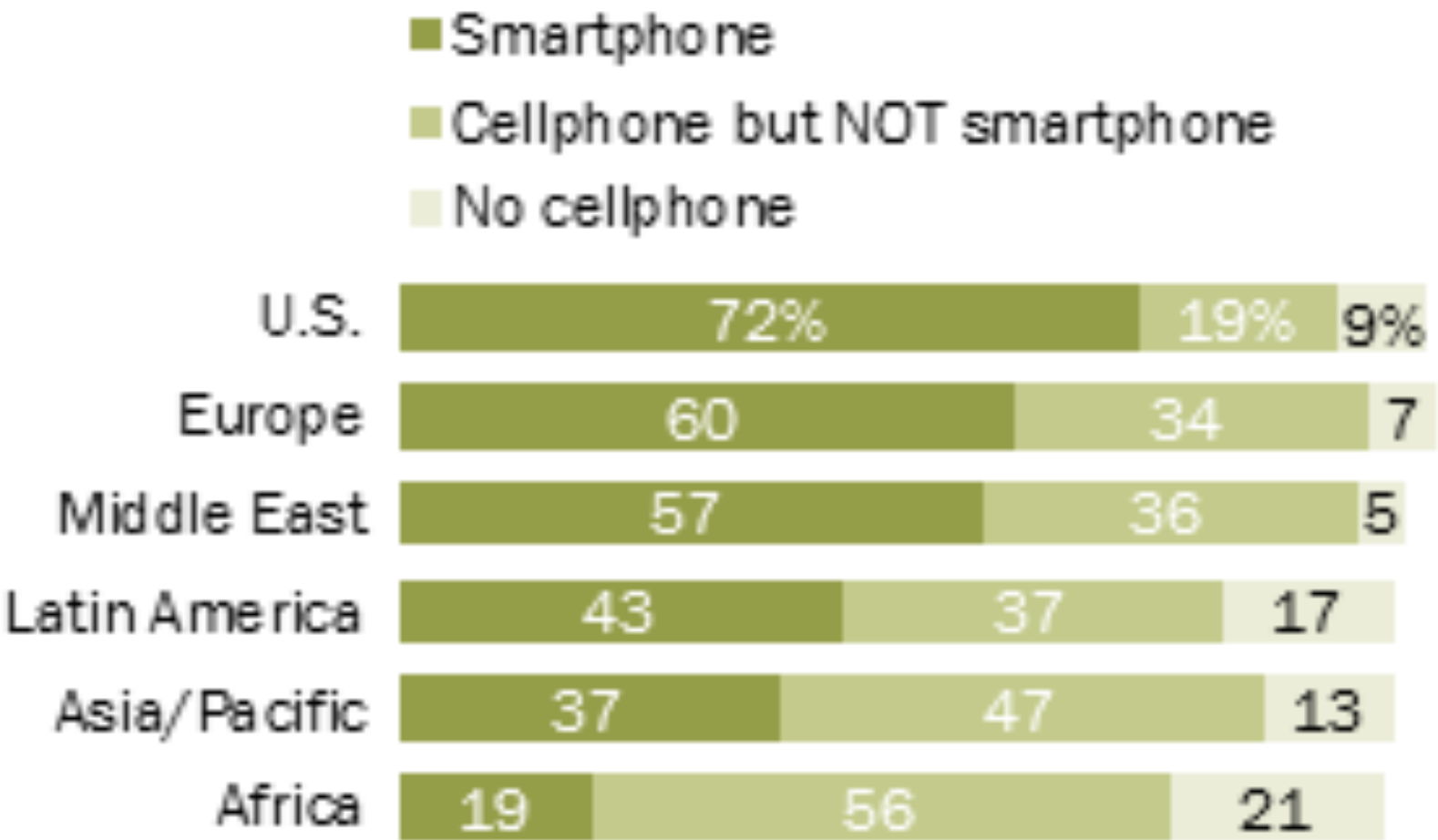
- ▶ Serve for different purposes
 - ▶ Composition: e.g., see for a discrete dimension x_i , the fraction of each values
 - ▶ Distribution: e.g., see the distribution of a continuous dimension of data x_i
 - ▶ Comparison:
 - ▶ Compare values of two continuous dimensions of the data, x_i and x_j
 - ▶ Given discrete x_i , compare the values of x_j when x_i takes different values.
 - ▶ Relationship: e.g., examine the relationship between x_i and x_j

COMPOSITION

- ▶ Pie charts
- ▶ Stacked bars
- ▶ Temporal trends
- ▶ Compare across groups

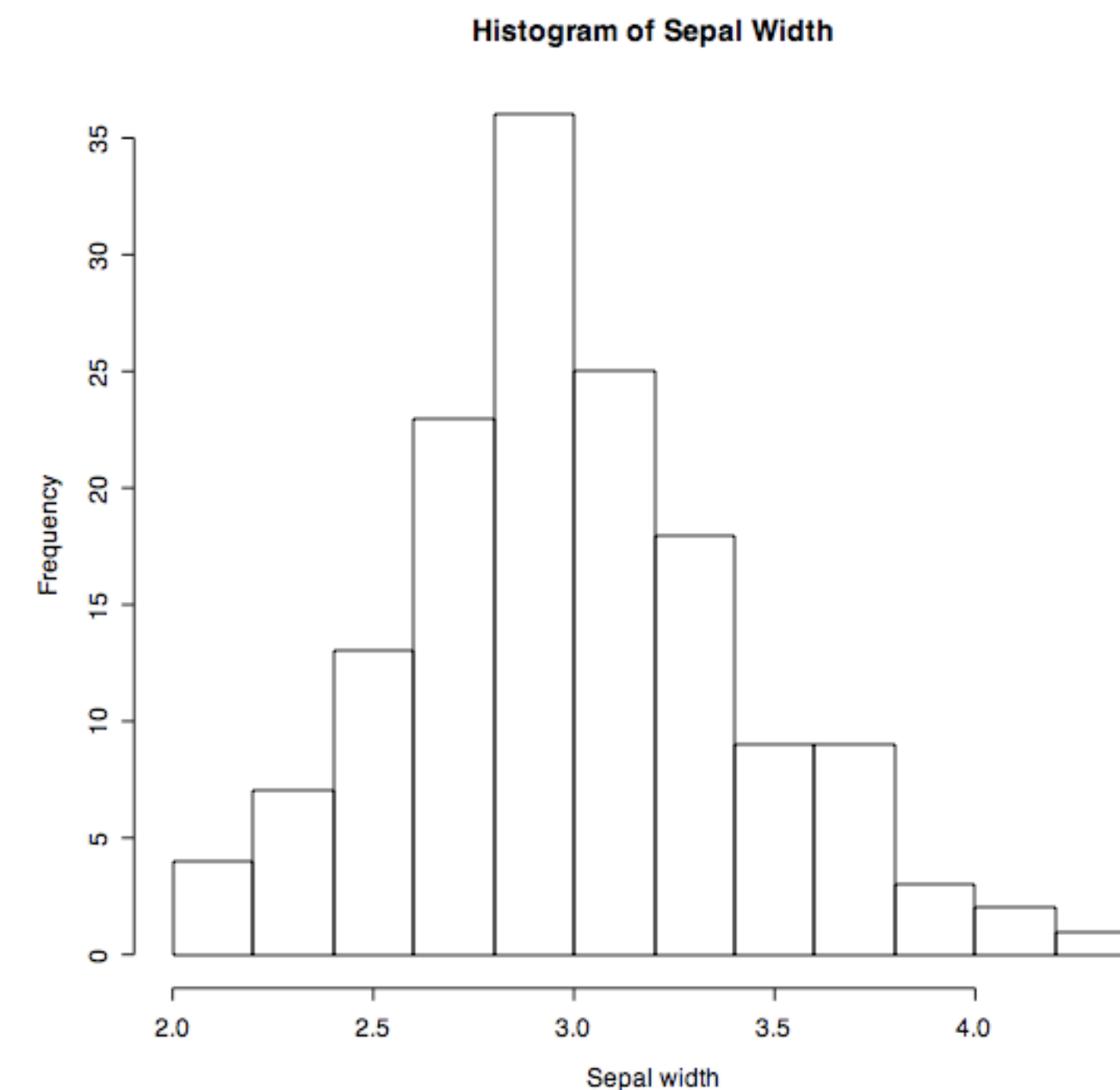


Regional medians of adults who report owning a ...



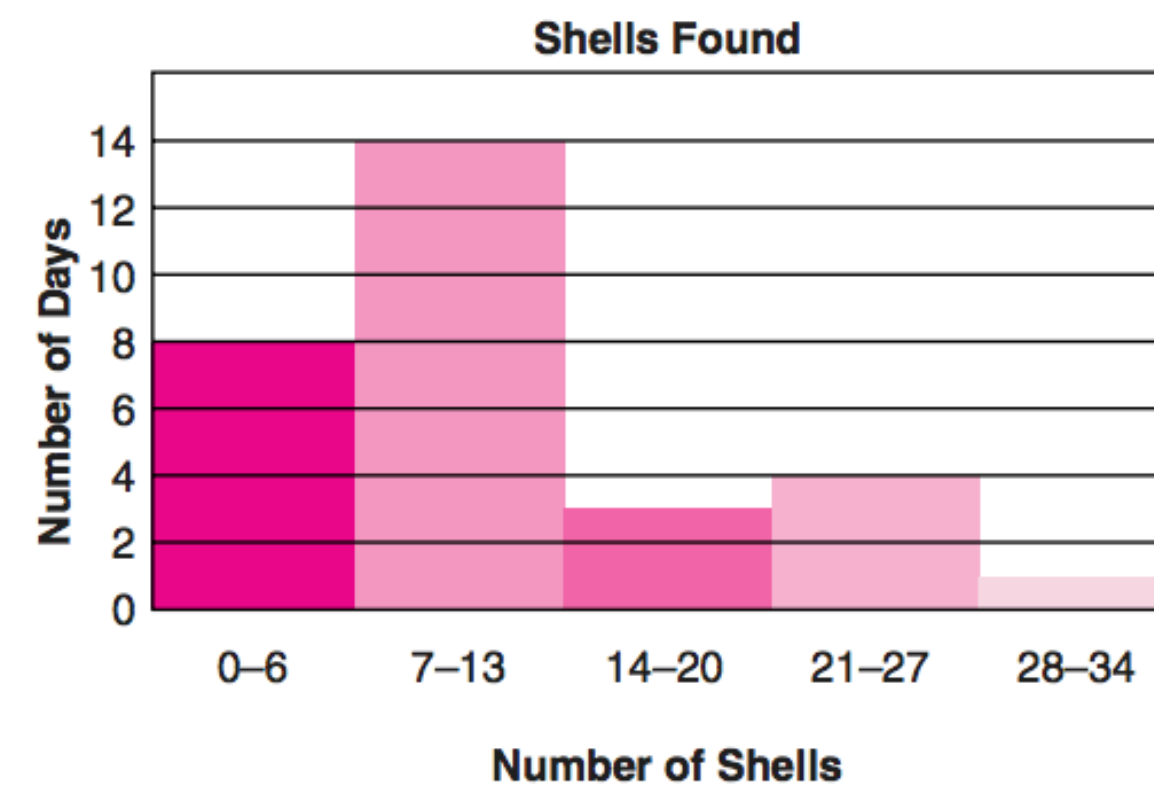
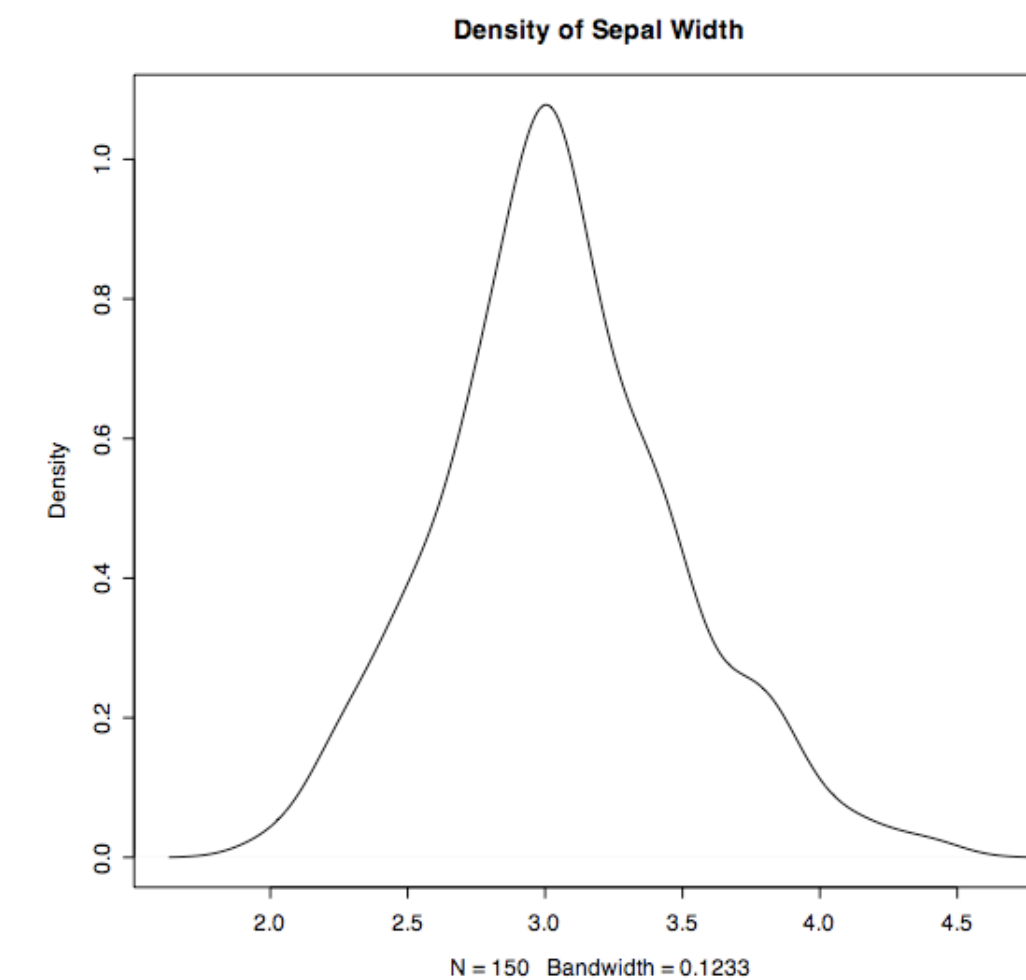
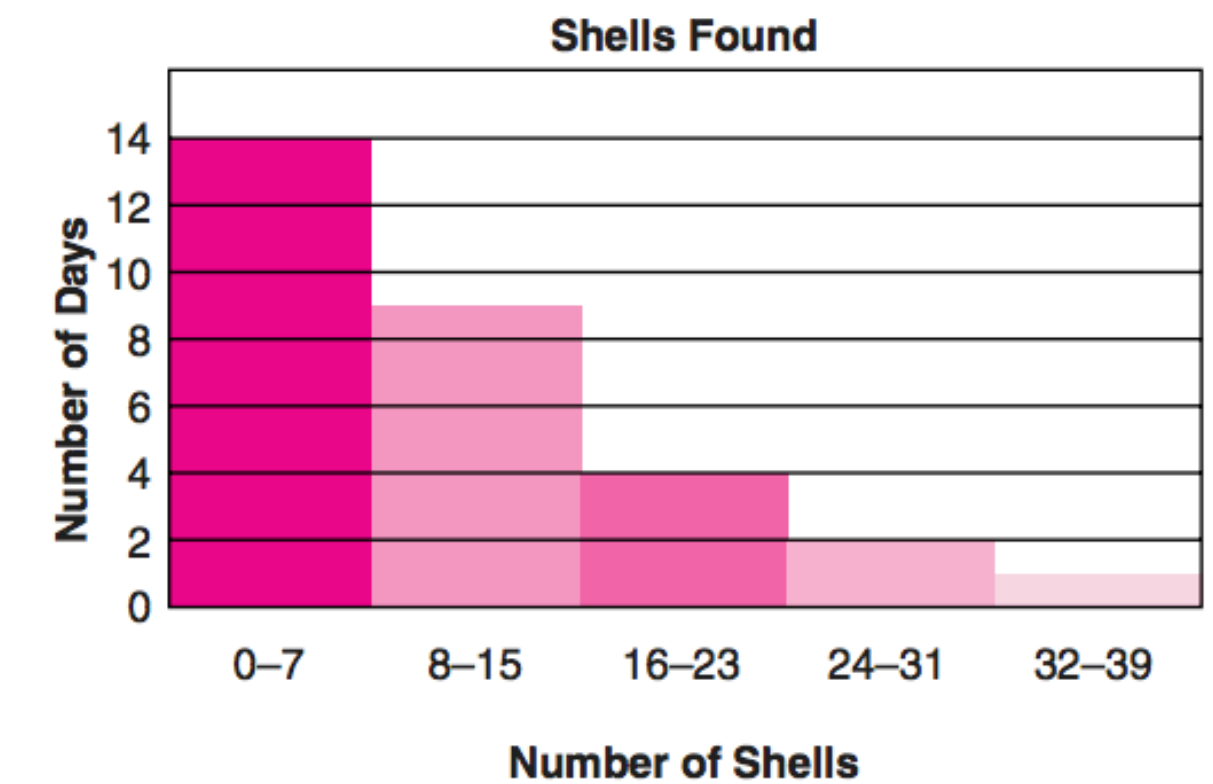
DISTRIBUTION: HISTOGRAMS (1D)

- ▶ Most common plot for univariate data
- ▶ Split data range into equal-sized bins, count number of data points that fall into each bin
- ▶ Graphically shows:
 - ▶ Center (location)
 - ▶ Spread (scale)
 - ▶ Skew
 - ▶ Outliers
 - ▶ Multiple modes



HISTOGRAM LIMITATIONS

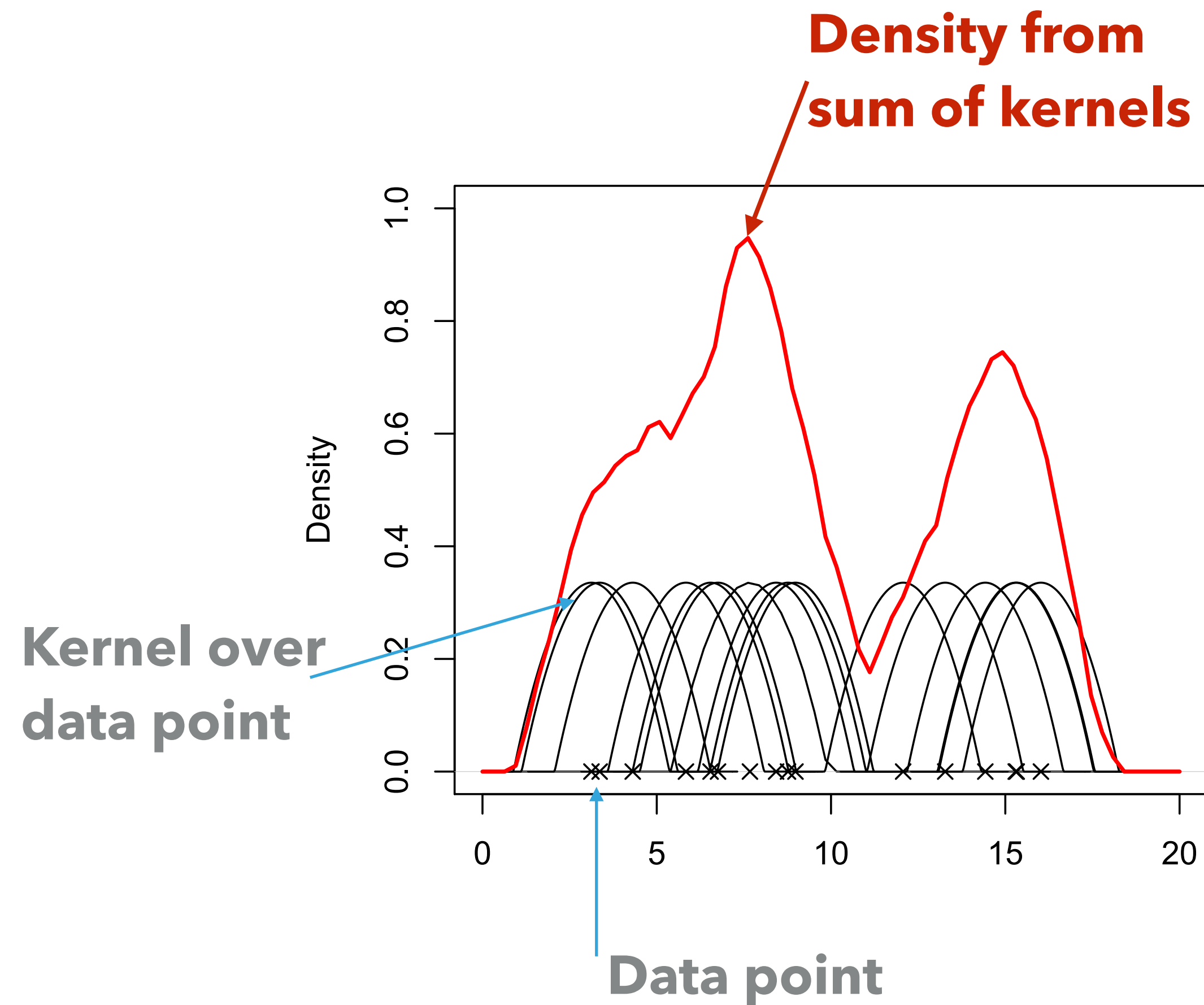
- ▶ Histograms can be misleading for small datasets
 - ▶ Slight changes in the data or binning approach can result in different histograms
- ▶ Solution: smoothed density plots
 - ▶ Use kernel function to estimate density at each point x , pools information from neighboring points

1.**2.**

KERNEL DENSITY ESTIMATION

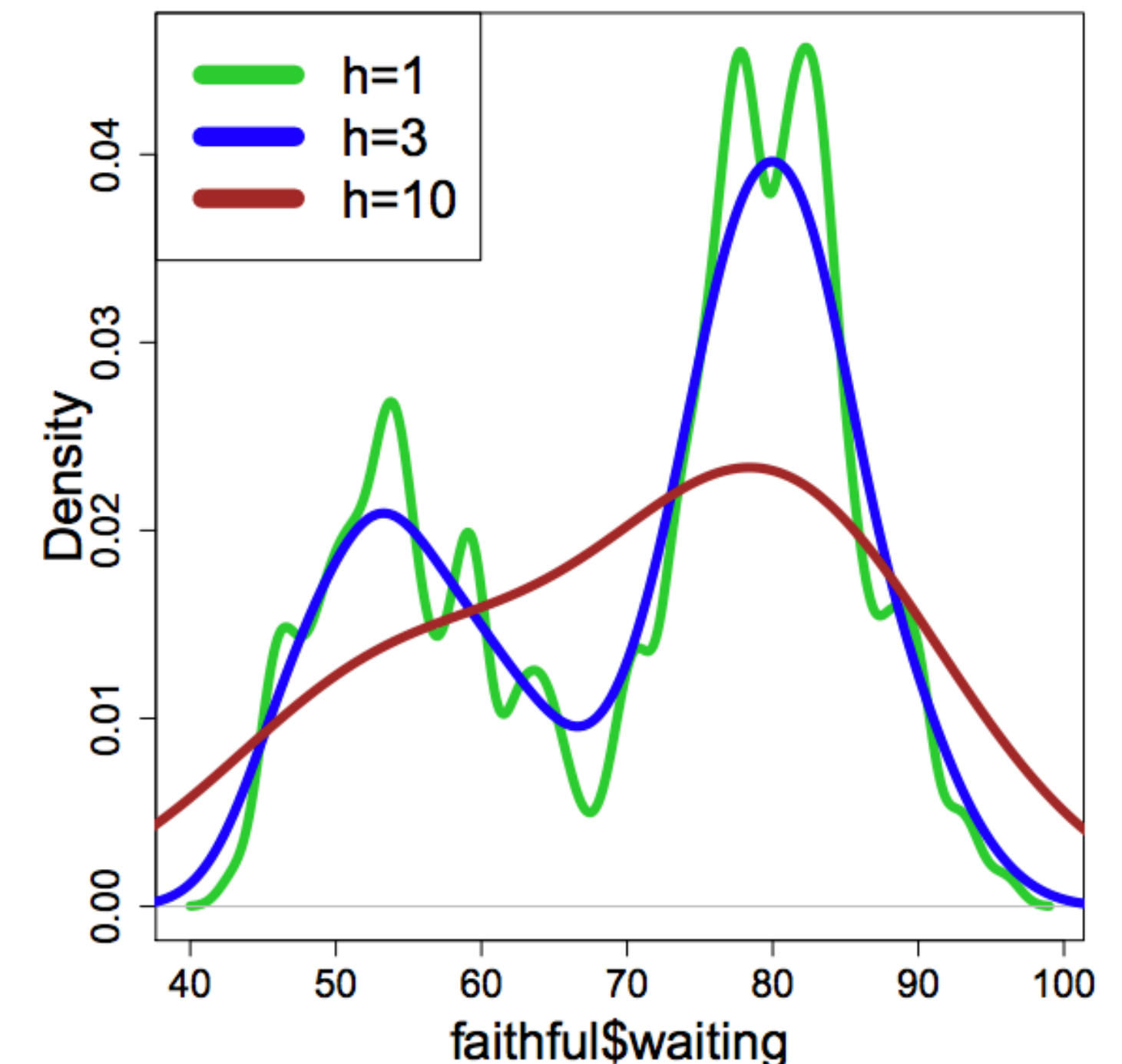
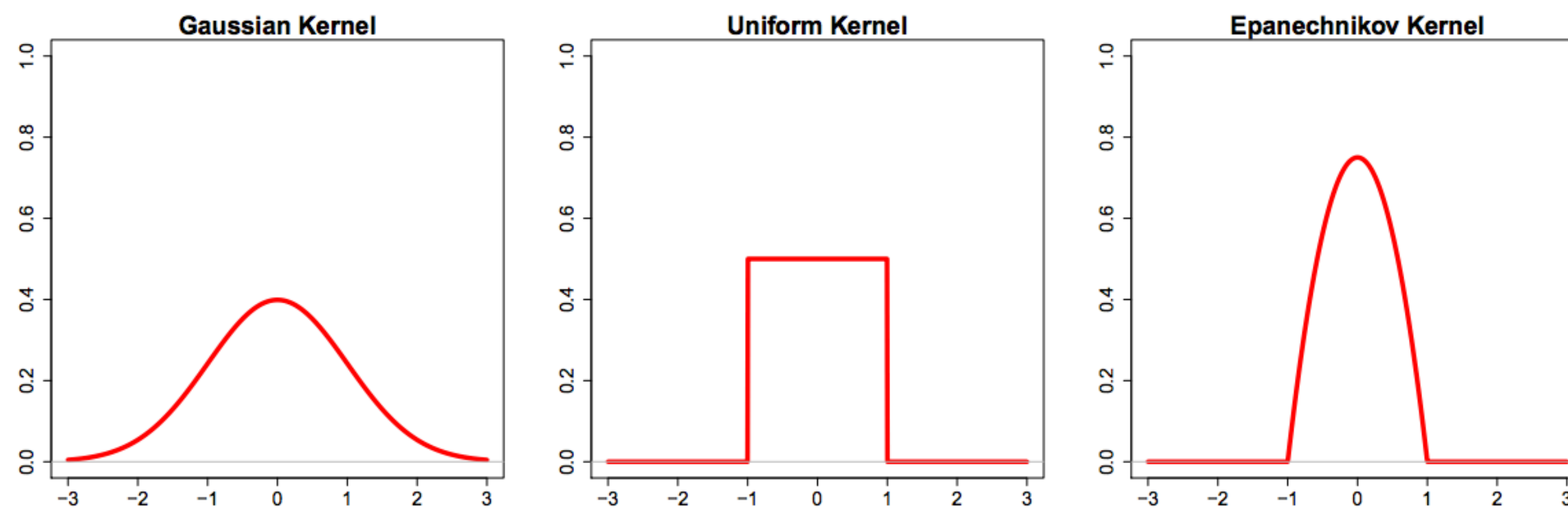
- Estimated density is:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^n K \left(\frac{x - x(i)}{h} \right)$$



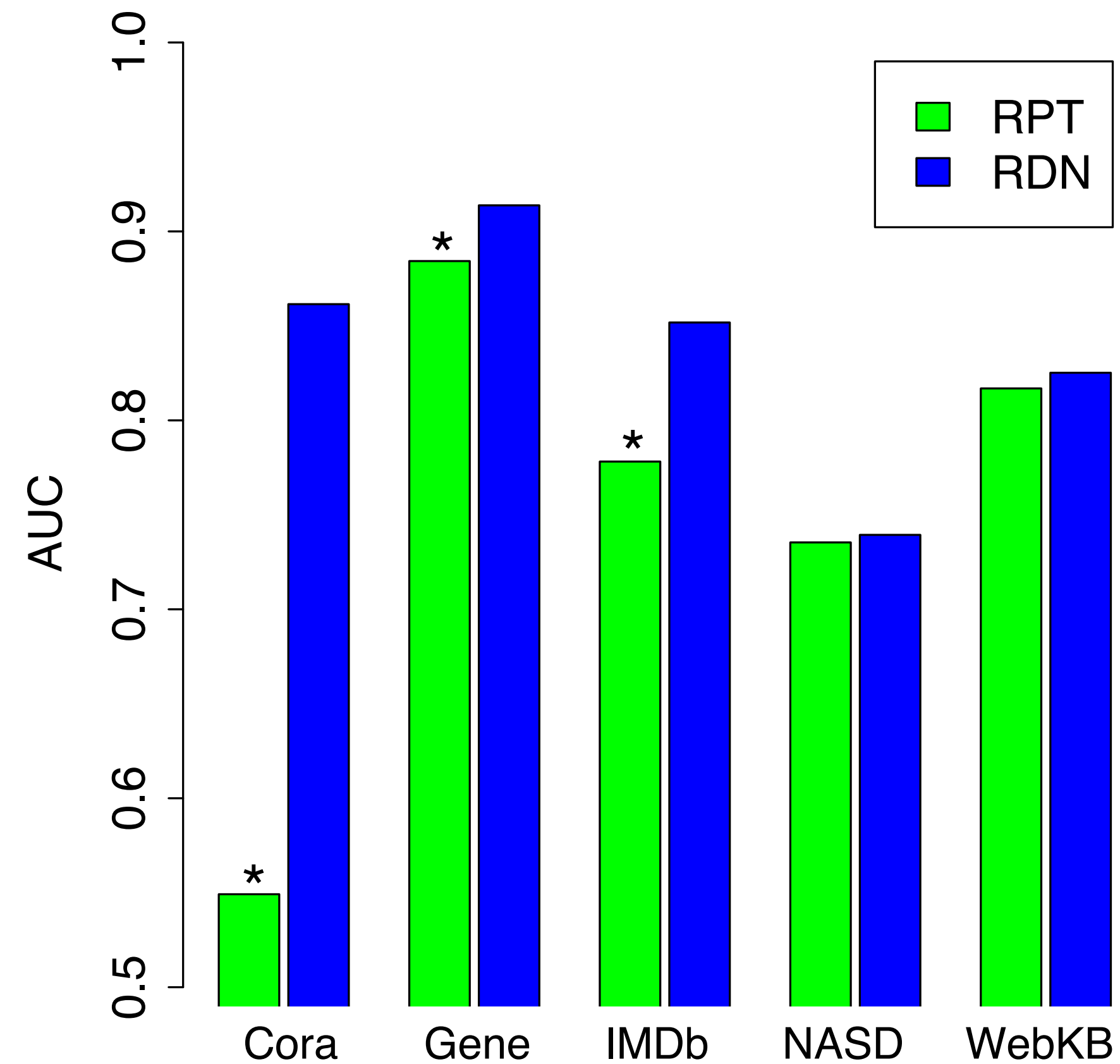
KERNEL DENSITY ESTIMATION

- ▶ Two parameters:
 - ▶ Kernel function K (e.g., Gaussian, Epanechnikov)
 - ▶ Bandwidth h



COMPARISON: BAR PLOTS

- ▶ Compare values of two continuous dimensions of the data, x_i and x_j



COMPARISON: BOX PLOT (2D)

- ▶ Display relationship between discrete and continuous variables
- ▶ For each discrete value X , calculate quartiles and range of associated Y values

